Name:	
Student ID:	

Midterm Examination - 2

Phys332: Electromagnetic Theory II

2026/04/15

Please carefully read below before proceeding!

I acknowledge by taking this examination that I am aware of all academic honesty conducts that govern this course and how they also apply for this examination. I therefore accept that I will not engage in any form of academic dishonesty including but not limited to cheating or plagiarism. I waive any right to a future claim as to have not been informed in these matters because I have read the syllabus along with the academic integrity information presented therein.

I also understand and agree with the following conditions:

(1) all calculations are to be conducted in the notations and conventions of the formulae sheets provided during the exam unless explicitly stated otherwise in the question;

(2) I take *full responsibility* for any ambiguity in my selections in "multiple choice questions";

(3) incorrect selections will receive $-1/7$ of the question's points;

(4) I am expected to provide *step-by-step explanation of how I solved the question* and am expected to do so *only within the answer boxes* provided with the questions: the explanation is supposed to be succinct, well-articulated, and correct both scientifically and mathematically;

(5) no partial credit is awarded for the explanations provided in the answer boxes;

(6) some questions of some students will be randomly selected for inspection: *a question (if selected for inspection) might be awarded negative points* if its explanation is incorrect or insufficient to get the correct answer, even if the correct option is selected;

(7) any page which does not contain *both my name and student id* may not be graded;

(8) any extra sheet that I may use are for my own calculations and will not be graded.

Signature: _____

Question:	1	2	3	4	Total
Points:	31½	31½	21	21	105

Name:	
Student ID:	

All dimensionfull quantities are to be understood in SI units.

(For instance, $\epsilon_0 \simeq 10^{-11}$ and $|\mathbf{B}| = 5$ stand for $\epsilon_0 \simeq 10^{-11}\,\mathrm{A}^2\,\mathrm{kg}^{-1}\,\mathrm{m}^{-3}\,\mathrm{s}^4$ and $|\mathbf{B}| = 5\,\mathrm{A}^{-1}\,\mathrm{kg}^1\,\mathrm{s}^{-2}$.)

Question: 1: One of the Midterm-1 Questions Reloaded ..... *(31½ points)*
(SOME HINT)

I will design a question almost identical to one in the first midterm, but with certain quantities (such as numerical values) changed.

.....
.....
.....
bla bla bla bla
.....
.....
.....

Please provide below the explanation of how you obtained your result(s) for question 1:

Question: 2: One of the Midterm-1 Questions Expanded ..... *(31½ points)*
(SOME HINT)

I will design a question as an expansion to what was already covered in a question of midterm one.

.....
.....
.....
bla bla bla bla
.....
.....
.....

Please provide below the explanation of how you obtained your result(s) for question 2:

Name:	
Student ID:	

Question: 3: A non-standard case that we have covered in class ..... *(21 points)*
(SOME HINT)

I will design a question focusing on the derivation of the relevant wave differential equation for a non-standard case (non-homogeneous, non-isotropic, non-linear, non-local, etc.) that we have covered in class.

.....
.....
.....
bla bla bla bla
.....
.....
.....

Please provide below the explanation of how you obtained your result(s) for question 3:

Question: 4: An explicit computation ..... *(21 points)*
(SOME HINT)

I will design a question requiring an explicit computation (something covered in book, such as attenuation in ohmic medium or reflection from a conductor) but probably from a different angle.

.....
.....
.....
bla bla bla bla
.....
.....
.....

Please provide below the explanation of how you obtained your result(s) for question 4:

« « « Congratulations, you have made it to the end, and possibly have realized that this document was prepared as part of an April Fool's trick! » » »